

Gamified Personal Tracker System - Professional Design Document

1. Vision

A scalable personal analytics and gamification system that tracks user activities and transforms them into an RPG-style progression system.

2. System Architecture

- 1 Frontend: React / Next.js
- 2 Backend: Spring Boot (Java 17)
- 3 Database: PostgreSQL
- 4 Cache: Redis
- 5 Future: Microservices architecture

3. Data Flow

- 1 Start activity → log entry created
- 2 Stop activity → duration calculated
- 3 XP calculated
- 4 Skill updated
- 5 Dashboard updated

4. Database Design

- 1 User(id, name, email)
- 2 Activity(id, name, category)
- 3 LogEntry(id, user_id, activity_id, start_time, end_time)
- 4 Skill(id, user_id, category, xp, level)
- 5 XPLog(id, log_entry_id, xp_earned)
- 6 Achievement(id, name, condition)

5. Gamification Engine

$XP = \text{duration_minutes} * \text{multiplier}$

$XP_required = 100 * (\text{level} ^ 1.5)$

6. API Design

- 1 POST /api/logs/start
- 2 POST /api/logs/stop
- 3 GET /api/dashboard
- 4 GET /api/skills

7. UI/UX

- 1 Dashboard with skill bars
- 2 Activity tracker buttons
- 3 Progress visualization

8. Deployment

- 1 Docker
- 2 AWS EC2
- 3 RDS PostgreSQL
- 4 CI/CD pipeline

9. Future Enhancements

- 1 AI insights
- 2 Quests system
- 3 Leaderboards
- 4 Mobile app